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LAMINATED HEADER FOR A MICROCHANNEL HEAT EXCHANGER

Abstract

A laminated header for a microchannel heat exchanger comprises a distribution plate comprising a first cut-out section of a first shape extending along a longitudinal axis of the distribution plate, wherein a width of the first cut-out section at least partially decreases along the longitudinal axis a hole plate comprising a plurality of bore-holes formed therein and separated by a distance therebetween, a section plate comprising a plurality of second cut-out sections of a second shape different from the first shape, and wherein the hole plate is parallelly stacked between the distribution plate and the section plate. The first cut-out section fluidically connects with each of the bore-holes and the second cut-out sections fluidically connects the bore-holes to a plurality of microchannel tubes of the heat exchanger.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This patent application claims the benefit of U.S. Provisional Patent Application No. 63/563,529, filed on Mar. 11, 2024, which is incorporated by reference herein in its entirety.

BACKGROUND

[0002] The subject disclosure relates to the field of heat exchangers, and more particularly, to a laminated header for a microchannel heat exchanger.

SUMMARY

[0003] Described herein is a laminated header for a microchannel heat exchanger. The header comprises, a distribution plate comprising a first cut-out section of a first shape extending along a longitudinal axis of the distribution plate, where a width of the first cut-out section at least partially along the longitudinal axis, a hole plate comprising a plurality of bore-holes formed therein and separated by a distance therebetween, and a section plate comprising a plurality of second cut-out sections of a second shape different from the first shape. The hole plate is parallelly stacked between the distribution plate and the section plate.

[0004] In one or more embodiments, the laminated header further comprises an inlet plate having an inlet bore-hole defined at a bottom end of the inlet plate, and wherein the width of the first cut-out section decreases from a bottom end towards a top end of the distribution plate.

[0005] In one or more embodiments, the first cut-out section has a trapezoidal shape with the width decreasing from the bottom end towards the top end of the distribution plate.

[0006] In one or more embodiments, the laminated header further comprises an inlet plate having an inlet bore-hole defined at a middle portion of the inlet, and wherein the width of the first cut-out section decreases from the middle portion towards a bottom end and a top end of the distribution plate.

[0007] In one or more embodiments, the laminated header further comprises a tube plate comprising a plurality of slots, wherein the laminated header is configured to receive and secure, within the plurality of slots, an end of a plurality of microchannel tubes of the microchannel heat exchanger.

[0008] In one or more embodiments, the distribution plate, the hole plate, and the section plate, are parallelly stacked between an inlet plate and a tube plate, and brazed together to form the laminated header, and wherein an end of the plurality of microchannel tubes of the heat exchanger is inserted within a plurality of slots of the tube plate and brazed to the laminated header to create a leak-proof connection between the laminated header and the microchannel tubes.

[0009] In one or more embodiments, the laminated header forms a fluidic passage that extends from the inlet bore-hole towards the plurality of slots while extending in a first direction along the longitudinal axis, between a top end and a bottom end, of the distribution plate and further extending from the first cut-out section into each of the second cut-out sections and the plurality of slots, via the plurality of bore-holes, in a second direction substantially perpendicular to the first direction and extending towards the plurality of slots or the microchannel tubes.

[0010] In one or more embodiments, the laminated header further comprising an inlet plate having an inlet bore-hole defined at a top end of the inlet plate, and wherein the width of the first cut-out section decreases from a top end towards a bottom end of the distribution plate.

[0011] In one or more embodiments, size or radii of the plurality of bore-holes associated with the

hole plate increases in the direction away from the inlet bore-hole.

[0012] In one or more embodiments, the inlet bore-hole is at a bottom end of the inlet plate, and wherein the size or radii of the plurality of bore-holes increases from a bottom end towards a top end of the hole plate.

[0013] In one or more embodiments, the inlet bore-hole is at a substantially middle portion of the inlet plate, and wherein the size or radii of the plurality of bore-holes increases from the middle portion towards a bottom end and a top end of the hole plate.

[0014] In one or more embodiments, the inlet bore-hole is at a top end of the inlet plate, and wherein the size or radii of the plurality of bore-holes increases from a top end towards a bottom end of the hole plate.

[0015] In one or more embodiments, the plurality of second cut-out sections associated with the section plate has a substantially rectangular or square profile.

[0016] In one or more embodiments, size of the plurality of second cut-out sections increases in at least one direction along the longitudinal axis.

[0017] In one or more embodiments, the distribution plate has a thickness that is greater than a thickness of the inlet plate and the hole plate.

[0018] In one or more embodiments, the laminated header is configured to allow a fluid from an inlet bore-hole defined on an inlet plate to flow within the first cut-out section of the distribution plate in along the longitudinal axis, between a top end and a bottom end, of the distribution plate.

[0019] In one or more embodiments, the header is further configured to allow uniform flow of the fluid from the first cut-out section into each of the second cut-out sections of the section plate, via the plurality of bore-holes of the hole plate, in a second direction perpendicular to the first direction and extending towards the plurality of microchannel tubes, and wherein the header further allows uniform flow of the fluid from the plurality of second cut-out sections into one or more ports associated with each of the microchannel tubes via the slots of the tube plate.

[0020] In one or more embodiments, the adjacent second cut-out sections among the plurality of second cut-out sections of the section plate are separated by a baffle extending in a direction, orthogonal to the first direction and the second direction, along a width of the section plate.

[0021] In one or more embodiments, the distribution plate, the hole plate, and the section plate are parallelly stacked between an inlet plate and a tube plate, and wherein, an inlet bore-hole of the inlet plate is formed at a corresponding widthwise center of the inlet plate, the plurality of bore-holes are formed at a corresponding widthwise center of the hole plate, and a plurality of slots of the tube plate are formed at a corresponding width-wise center of a width of a tube plate.

[0022] Also described herein is a heat exchanger comprises the one or more laminated headers; wherein the one or more headers are coaxially and sequentially stacked in a longitudinal direction to form a vertical header having one or more compartments, wherein the plurality of microchannel tubes associated with the heat exchanger is fluidically connected to the plurality of slots associated with the one or more headers.

[0023] In one or more embodiments, the heat exchanger comprises an external distributor comprising: an inlet configured to be fluidically connected to a supply tube; and one or more outlets, each configured to be fluidically connected to one of the laminated headers via feeder tubes or tube stubs.

[0024] A further aspect of the subject disclosure relates to a laminated header for a microchannel heat exchanger. The laminated header comprises, a distribution plate comprising a first cut-out section of a first shape, a hole plate comprising a plurality of bore-holes formed therein and separated by a distance therebetween, a diameter of each of the plurality of bore-holes varies along a longitudinal axis of the hole plate, and a section plate comprising a plurality of second cut-out sections of a second shape different from the first shape, wherein the hole plate is parallelly stacked between the distribution plate and the section plate.

[0025] The foregoing summary is illustrative only and is not intended to be in any way limiting. In

addition to the illustrative aspects, embodiments, and features described above, further aspects, embodiments, features, and techniques of the subject disclosure will become more apparent from the following description taken in conjunction with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0026] The accompanying drawings are included to provide a further understanding of the subject disclosure and are incorporated in and constitute a part of this specification. The drawings illustrate exemplary embodiments of the subject disclosure and, together with the description, serve to explain the principles of the subject disclosure.

[0027] In the drawings, similar components and/or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label with a second label that distinguishes among the similar components. If only the first reference label is used in the specification, the description is applicable to any one of the similar components having the same first reference label irrespective of the second reference label.

[0028] FIG. 1A illustrates exemplary side cross-sectional side view of an embodiment of the laminated vertical header, in accordance with one or more embodiments of the subject disclosure.

[0029] FIG. 1B illustrates an exemplary front view of an inlet plate associated with the header of FIG. 1A, in accordance with one or more embodiments of the subject disclosure.

[0030] FIG. 1C illustrates an exemplary front view of a distribution plate associated with the header of FIG. 1A, in accordance with one or more embodiments of the subject disclosure.

[0031] FIG. 1D illustrates an exemplary front view of a hole plate associated with the header of FIG. 1A, in accordance with one or more embodiments of the subject disclosure.

[0032] FIG. 1E illustrates an exemplary front view of a section plate associated with the header of FIG. 1A, in accordance with one or more embodiments of the subject disclosure.

[0033] FIG. 1F illustrates an exemplary front view of a tube plate associated with the header of FIG. 1A, in accordance with one or more embodiments of the subject disclosure.

[0034] FIG. 2A illustrate exemplary side cross-sectional side view of another embodiment of the laminated vertical header, in accordance with one or more embodiments of the subject disclosure.

[0035] FIG. 2B illustrates an exemplary front view of the inlet plate associated with the header of FIG. 2A, in accordance with one or more embodiments of the subject disclosure.

[0036] FIG. 2C illustrates an exemplary front view of a distribution plate associated with the header of FIG. 2A, in accordance with one or more embodiments of the subject disclosure.

[0037] FIG. 2D illustrates an exemplary front view of a hole plate associated with the header of FIG. 2A, in accordance with one or more embodiments of the subject disclosure.

[0038] FIG. 2E illustrates an exemplary front view of a section plate associated with the header of FIG. 2A, in accordance with one or more embodiments of the subject disclosure.

[0039] FIG. 2F illustrates an exemplary front view of a tube plate associated with the header of FIG. 2A, in accordance with one or more embodiments of the subject disclosure.

[0040] FIG. 3 illustrates exemplary side cross-sectional side view depicting multiple laminated vertical headers stacked together and connected to form a single external fluid distributor, in accordance with one or more embodiments of the subject disclosure.

[0041] FIG. 4A illustrates an exemplary front view of the distribution plate associated with the header of FIGS. 1A and 2A where the inlet bore-hole is at middle portion of the first plate, in accordance with one or more embodiments of the subject disclosure.

[0042] FIG. 4B illustrates an exemplary front view of the inlet plate associated with the header of FIGS. 1A and 2A where the inlet bore-hole is at middle portion of the first plate, in accordance with one or more embodiments of the subject disclosure.

[0043] FIG. 4C illustrates an exemplary front view of the distribution plate associated with the header of FIGS. 1A and 2A where the inlet bore-hole is at top end of the first plate, in accordance with one or more embodiments of the subject disclosure.

[0044] FIG. 4D illustrates an exemplary front view of the inlet plate associated with the header of FIGS. 1A and 2A where the inlet bore-hole is at top end of the first plate, in accordance with one or more embodiments of the subject disclosure.

[0045] FIG. 5A illustrates an exemplary front view of a first plate associated with yet another embodiment of the laminated vertical header where a portion of the first cut-out section, between the inlet bore-hole and the bore-hole adjacent to the inlet bore-hole, of an distribution plate has a substantially narrow passage or area, in accordance with one or more embodiments of the subject disclosure.

[0046] FIG. 5B illustrates an exemplary front view of the distribution plate associated with the header of FIG. 5A, in accordance with one or more embodiments of the subject disclosure.

[0047] FIG. 5C illustrates an exemplary front view of a hole plate associated with the header of FIG. 5A, in accordance with one or more embodiments of the subject disclosure.

[0048] FIG. 5D illustrates an exemplary front view of a section plate associated with the header of FIG. 5A, in accordance with one or more embodiments of the subject disclosure.

[0049] FIG. 5E illustrates an exemplary front view of a tube plate associated with the header of FIG. 5A, in accordance with one or more embodiments of the subject disclosure.

[0050] FIG. 6A illustrates an exemplary front view of a first plate associated with another embodiment of the laminated vertical header where the inlet bore-hole is at middle portion of the first plate and a portion of the first cut-out section, between the inlet bore-hole and the bore-hole adjacent to the inlet bore-hole, has a substantially narrow passage or area, in accordance with one or more embodiments of the subject disclosure.

[0051] FIG. 6B illustrates an exemplary front view of the distribution plate associated with the header of FIG. 6A, in accordance with one or more embodiments of the subject disclosure.

[0052] FIG. 6C illustrates an exemplary front view of a hole plate associated with the header of FIG. 6A, in accordance with one or more embodiments of the subject disclosure.

[0053] FIG. 6D illustrates an exemplary front view of a section plate associated with the header of FIG. 6A, in accordance with one or more embodiments of the subject disclosure.

[0054] FIG. 6E illustrates an exemplary front view of a tube plate associated with the header of FIG. 5A, in accordance with one or more embodiments of the subject disclosure.

DETAILED DESCRIPTION

[0055] The following is a detailed description of embodiments of the subject disclosure depicted in the accompanying drawings. The embodiments are in such detail as to clearly communicate the subject disclosure. However, the amount of detail offered is not intended to limit the anticipated variations of embodiments; on the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the subject disclosure as defined by the appended claims.

[0056] Various terms are used herein. To the extent a term used in a claim is not defined below, it should be given the broadest definition persons in the pertinent art have given that term as reflected in printed publications and issued patents at the time of filing.

[0057] In the specification, reference may be made to the spatial relationships between various components and to the spatial orientation of various aspects of components as the devices are depicted in the attached drawings. However, as will be recognized by those skilled in the art after a complete reading of the subject disclosure, the components described herein may be positioned in any desired orientation. Thus, the use of terms such as “above,” “below,” “upper,” “lower,” “first,” “second” or other like terms to describe a spatial relationship between various components or to describe the spatial orientation of aspects of such components should be understood to describe a relative relationship between the components or a spatial orientation of aspects of such

components, respectively, described herein may be oriented in any desired direction.

[0058] Microchannel heat exchangers (MCHX) are widely employed in heating, ventilation, and air conditioning (HVAC) systems for their compact size, high heat transfer efficiency, and improved energy efficiency. However, the effectiveness of heat transfer in the MCHX may impact the overall efficiency of the HVAC system. One of the challenges faced in MCHX is maldistribution of two-phase flow at the inlet of heat exchange tubes associated with the MCHX. Maldistribution occurs when there is an uneven distribution of refrigerant between the tubes, leading to variations in local heat transfer rates. In particular, maldistribution within a vertical header of the MCHX poses a substantial hurdle, as the force of gravity acts against maintaining a consistent mixing of liquid and vapor along the header's height. There is, therefore, a need to provide an improved vertical header for MCHX which enables uniform supply of fluid (refrigerant) into the ports of the heat exchange tubes connected to the vertical header while restricting separation of liquid-phase within the header under gravity.

[0059] Referring to FIGS. 1A to 2F, a laminated header **100** (referred to as header **100**, hereinafter) for a heat exchanger is disclosed. The header **100** may include a first plate **102** (also referred to as an inlet plate **102**, herein), that further includes a first/inlet bore-hole **102-1** formed therein. Further, the header **100** may include a second plate **104** (also referred to as a distribution plate **104**, herein) that includes a first cut-out section **104-1** of a first shape extending along a longitudinal axis of the second plate **104**. The header **100** may further include a third plate **106** (also referred to as a hole plate, herein) that includes a plurality of (second) bore-holes **106-1** formed therein and separated by a distance therebetween. In addition, the header **100** may include a fourth plate **108** (also referred to as a section plate, herein) including a plurality of second cut-out sections **108-1** of a second shape. Further, the header **100** may include a fifth plate **110** (also referred to as a tube plate, herein) that includes a plurality of slots S. The detailed shape and construction of these plates have been explained in conjunction with FIGS. 1B to 1F and 2B to 2F.

[0060] In one or more embodiments, the second plate **104** may be parallelly stacked between the first plate **102** and the third plate **106**, and the fourth plate **108** may be parallelly stacked between the third plate **106** and the fifth plate **110** in same orientation to form the (stacked) header **100**, such that the first cut-out section **104-1** fluidically connects the inlet bore-hole **102-1** to each of the bore-holes **106-1** and the second cut-out sections **108-1** fluidically connects the bore-holes **106-1** to the slots S. Further, the (stacked) header **100** may be configured to receive and secure, within the plurality of slots S, an end of a plurality of microchannel tubes **112** associated with the heat exchanger, thereby fluidically connecting the inlet bore-hole **102-1** to the microchannel tubes **112**. The size of slots S may be based on size of individual microchannel tubes **112**, such that the ends of the microchannel tubes **112** can be securely fitted in the corresponding slots S.

[0061] In one or more embodiments, the first plate **102**, the second plate **104**, the third plate **106**, the fourth plate **108**, and the fifth plate **110** may be parallelly stacked and brazed together to form the laminated header **100**. Further, the end of the microchannel tubes **112** may be inserted within the slots S of the laminated header **100** and further brazed to the laminated header **100** to create a leak-proof connection between the laminated header **100** and the microchannel tubes **112**.

Accordingly, the (stacked) header **100** forms a fluidic passage that extends from the inlet bore-hole **102-1** of the first plate **102** towards the plurality of slots S of the third plate **106** while extending in a first direction (A) along a length/longitudinal axis, between a top end and a bottom end, of the second plate **104** and further extending from the first cut-out section **104-1** into each of the second cut-out sections **108-1** and the plurality of slots S or the microchannel tubes **112**, via the plurality of bore-holes **106-1**, in a second direction (B) substantially perpendicular to the first direction (A). In some embodiments, the longitudinal axis corresponds to the axis along the longest dimension of the (second) plates. In some embodiments, the longitudinal direction may correspond to a length or dimension from the bottom end to the top end of the second plate **104**, as shown in FIGS. 1A to 2F. In some embodiments, the longitudinal axis may be dimensions along which the microchannel

tubes **112** are arranged, or direction/axis along which the bore-holes are **106-1** are arranged aligned. Further, the first-bore hole of the first plate **102** may be fluidically connected to a supply tube **114** associated with the heat exchanger that may be configured to supply two-phase refrigerant (fluid) into the header **100**.

[0062] In one or more embodiments, the second plate **104**, the third plate **106**, and the further plate **108** may be stacked parallelly and brazed together. In such embodiments, the first cut-out section **104-1** of the second plate **104**, the bore-holes **106-1** of the hole plate **106**, and the second cut-out sections **108-1** of the section plate **108** may form a fluidical passage therebetween. In some embodiments, a fluid may be received from an inlet bore-hole, such as inlet bore-hole **102-1** on the inlet plate **102**, or any other inlet bore-hole defined on a cover that encapsulates the first cut-out section **104-1** of the second plate **104**. In some embodiments, the cover may be hermetically attached to the second plate **104**. The fluid may pass through the first cut-out section **104-1**, the bore-holes **106-1**, and the second cut-out sections **108-1**, as described subsequently in the subject disclosure. Further, the fluid flow into the microchannel tubes **112**, which may be fluidically connected to the second cut-out sections **108-1** through slots, such as slots **S** on the fifth plate **110**, or slots defined on a slotted plate. The slotted plate may be attached to the fourth plate **108**.

[0063] In one or more embodiments, as illustrated in FIGS. **1B** to **1F** and **2B** to **2F**, the inlet bore-hole **102-1** may be formed at a width-wise center (i.e., center with respect to the width) the first plate **102**. The bore-holes **106-1** may be formed at a width-wise center of the third plate **106** with a space therebetween. Further, the slots **S** may be formed at a width-wise center of the fifth plate **110**, each of the slots **S** being separated by a partition wall. Furthermore, in one or more embodiments, the second plate **104** may have the same or different thickness compared to the first plate **102** and the third plate **106** to facilitate flow of the fluid with minimal back pressure. However, in some embodiments, the first, second, and third plate **106s** may also have same thickness without any limitations.

[0064] In one or more embodiments, a width (i.e., a dimension perpendicular to the longitudinal axis, and/or direction of flow of a fluid through the laminated header **100**), and correspondingly volume of (or area defined by), the first cut-out section **104-1** may decrease along its length/longitudinal axis in at least one direction (i.e., in a direction away from) the inlet bore-hole **102-1**. This configuration can keep the same mass flow rate at the inlet of all bore-holes **106-1** of the third plate **106** and further into the ports of the of microchannel tubes **112**.

[0065] Referring to FIGS. **1B** to **1C**, in one or more embodiments, the inlet bore-hole **102-1** of the first plate **102** may be at a bottom end of the first plate **102**. Accordingly, the width (and correspondingly area or volume) of the first cut-out section **104-1** may decrease along a bottom end towards a top end of the second plate **104**. As illustrated, the first cut-out section **104-1** may have a trapezoidal shape/contour with the volume/width decreasing from the bottom end towards the top end of the second plate **104**. The decreasing volume/width may help ensure equal fluid flow velocity in the flow channel as the part of the two-phase mixture is fed through holes **106-1**.

[0066] Referring to FIGS. **4A** and **4B**, in one or more embodiments, the inlet bore-hole **102-1** of the first plate **102** may be at a substantially middle portion of the first plate **102**. Accordingly, the width (and correspondingly area or volume) of the first cut-out section **104-1** may decrease from the middle portion towards a bottom end and/or a top end of the second plate **104**. In such embodiments, the first cut-out section **104-1** may have a shape/contour similar to that of a stretched hexagon or a diamond.

[0067] Referring to FIGS. **4C** and **4D**, in one or more embodiments, the inlet bore-hole **102-1** of the first plate **102** may be at a top end of the first plate **102**. Accordingly, the width (and correspondingly area or volume) of the first cut-out section **104-1** may decrease from a top end towards a bottom end of the second plate **104**.

[0068] In one or more embodiments, size or radii of the bore-holes **106-1** associated with the third plate **106** may increase in at least one direction along the longitudinal axis (such as along the

direction away from the inlet bore-hole **102-1**). In such embodiments, the width/volume of the first cut-out section **104-1** may be equal or uniform along its length/longitudinal axis. The first cut-out section **104-1** may have a substantially rectangular or square profile. This configuration may keep the same or uniform mass flow rate at the outlet of all bore-holes **106-1** of the third plate **106** and further into the ports of the of microchannel tubes **112**.

[0069] Referring to FIGS. **2B** to **2D**, in one or more embodiments, the inlet bore-hole **102-1** may be at a bottom end of the first plate **102**. Accordingly, the size or radii of each of the bore-holes **106-1** may increase from a bottom end towards a top end of the third plate **106**. For example, the bore-hole **106-1** at or proximate to the bottom end of the third plate **106** may have a smaller size/diameter in comparison to the bore-hole **106-1** at or proximate to the top end of the third plate **106**. The bore-holes **106-1** may be arranged in increasing order of sizes/diameters from the bottom end to the top end of the third plate **106**.

[0070] Further, in one or more embodiments (as shown in FIGS. **4A** and **4B**), the inlet bore-hole **102-1** may be at a substantially middle portion of the first plate **102**. Accordingly, the size or radii of the plurality of bore-holes **106-1** may increase from the middle portion towards a bottom end and a top end of the third plate **106**. In such embodiment, the bore-holes **106-1** at or in proximity to the center of the third plate **106** may have the smallest radii or size of all the bore-holes **106-1**, and each successive bore-hole **106-1** defined from the center/middle to the top end and/or the bottom end of the third plate **106** may have a larger size or radii than the preceding second-bore hole.

[0071] Furthermore, in one or more embodiments (as shown in FIGS. **4C** and **4D**), the inlet bore-hole **102-1** may be at a top end of the first plate **102**. Accordingly, the size or radii of the plurality of bore-holes **106-1** may increase from a top end towards a bottom end of the third plate **106**.

[0072] Referring back to FIGS. **1B** to **1F** and **2B** to **2F**, in one or more embodiments, the plurality of second cut-out sections **108-1** associated with the fourth plate **108** may have a substantially rectangular or square profile. Further, in one or more embodiments (not shown), the size of the plurality of second cut-out sections **108-1** may increase along the longitudinal axis, i.e., in the direction away from the inlet bore-hole **102-1**. For example, the second cut-out sections **108-1** may be arranged in increasing order of sizes from the top end to the bottom end of the fourth plate **108** (when the inlet bore-hole **102-1** is defined on the bottom end of the first plate **102**), or from the bottom end to the top end of the fourth plate **108** (when the inlet bore-hole **102-1** defined on the top end of the first plate **102**). However, in some embodiments, the size of each of the second cut-out sections **108-1** may be equal and uniform along the longitudinal axis.

[0073] In one or more embodiments, the adjacent second cut-out sections **108-1** associated with the fourth plate **108** may be separated by a baffle **108-2** (shown in FIGS. **1B** to **1F** and **2B** to **2F**) extending in a direction (C), orthogonal to the first direction (A) and the second direction (B), along a width of the fourth plate **108**.

[0074] Accordingly, referring to FIGS. **1A** and **2A**, the (stacked) header **100** may be configured to receive a fluid via the inlet bore-hole **102-1** of the first plate **102** and further allow the flow of the received fluid within the first cut-out section **104-1** of the second plate **104** in a first direction (A) along a length/longitudinal axis, between a top end and a bottom end, of the second plate **104**. The header **100** may be further configured to allow uniform flow of the fluid from the first cut-out section **104-1** into each of the second cut-out sections **108-1** of the fourth plate **108**, via the bore-holes **106-1** of the third plate **106**, in a second direction (B) perpendicular to the first direction (A) and extending towards the plurality of microchannel tubes **112**. Furthermore, the header **100** may allow uniform flow of the fluid from the plurality of second cut-out sections **108-1** into one or more ports associated with each of the microchannel tubes **112** via the slots S of the fifth plate **110**.

[0075] Referring to FIGS. **5A** to **5E**, and **6A** to **6E**, in one or more embodiments, a constriction or a narrow passage/area is defined between the inlet bore-hole **102-1** and the bore-hole **106-1** adjacent to the inlet bore-hole **102-1**, on the first cut-out section **104-1** of the second plate **104** (also referred to as a constriction, herein). As illustrated in FIGS. **5A** to **5E**, in one or more embodiments, when

the inlet bore-hole **102-1** is at a bottom end of the first plate **102**, a portion P at a bottom end of the first cut-out section **104-1**, between the inlet bore-hole **102-1** and the bottom-most bore-hole **106-1** (adjacent to the inlet bore-hole **102-1**) may define the constriction, i.e., a substantially narrow passage or area (C). Further, as illustrated in FIGS. **6A** to **6E**, in one or more embodiments, when the inlet bore-hole **102-1** is in a middle of the first plate **102**, the portions **P1**, **P2** of the first cut-out section **104-1**, between the inlet bore-hole **102-1** and the bore-holes **106-1** (above and below the inlet bore-hole **102-1**) may define the constriction, i.e., a substantially narrow passage or area (C). In such embodiments, the first cut-out section **104-1** may have two narrowed portions **P1**, **P2** adjacent to the opposite ends of the inlet bore-hole **102-1**.

[0076] Referring to FIG. **3**, in one or more embodiments, a heat exchanger comprising one or more laminated headers **100** of FIGS. **1A** and/or **2A** is disclosed. As illustrated, the headers **100-1** to **100-N** may be coaxially and sequentially stacked in a longitudinal direction to form a single vertical header **300** having one or more compartments (headers **100-1** to **100-N**). Further, the plurality of microchannel tubes **112** associated with the heat exchanger may be fluidically connected to the plurality of slots **S** associated with the headers **100-1** to **100-N**.

[0077] In addition, in one or more embodiments, the heat exchanger may include an external distributor **302** having an inlet, and one or more outlets connected to the inlet via fluidic passages. The inlet of the distributor **302** may be configured to be fluidically connected to a supply tube **114**. Further, each of the outlets of the distributor **302** may be configured to be fluidically connected to the inlet bore-hole **102-1** associated with one of the headers **100-1** to **100-N** via feeder tubes **304** or tube stubs to provide two-phase flow with equal volumes at the inlet of each compartment formed in the corresponding header **100**. Accordingly, the distributor **302** may supply equal volume of fluid into each of the header **100-1** to **100-N** (or compartments of the vertical header **300**). Further, each of the header **100-1** to **100-N** (compartments) may be configured to uniformly supply an equal volume of the received fluid into ports of each of the microchannel tubes **112** of the heat exchanger.

[0078] It is to be appreciated that the decrease in the flow area of the first cut-out section **104-1** in the second plate **104** in a direction away from the inlet bore-hole **102-1** may allow the same mass flow rate of the fluid at the inlet of all holes of the third plate **106** and further into the ports of the of microchannel tubes **112**. Moreover, when the flow area of the first cut-out section **104-1** is uniform along its length/longitudinal axis in the second plate **104**, the increase in size or radii of the second-bore holes of the third plate **106** (while moving) along the longitudinal direction, such as in a direction away from the inlet bore-hole **102-1**, may allow the same mass flow rate of the fluid at the outlet of all holes of the third plate **106** and further into the ports of the of microchannel tubes **112**. Hence, the volume of the fluid may be uniformly distributed out of the third plate **106** and eventually to the microchannel tubes **112**, thereby preventing the problems associated with maldistribution of two-phase flow to the microchannel tubes **112** associated with the heat exchanger, such as variations in local heat transfer rates and maintaining of a consistent mixing of liquid and vapour along the header's height.

[0079] Thus, the subject disclosure overcomes the challenges associated with existing heat exchangers, by providing an improved laminated vertical header for the heat exchanger. The header uniformly supplies the fluid (refrigerant) into the ports of each of the tubes while maintaining a lower pressure drop and and restricting separation of liquid-phase within the header under gravity, thereby improving the performance and efficiency of the overall heat exchanger. In addition, the simple design of the header makes it easier to manufacture as well as cost-effective, and further allows these headers to be stacked coaxially to increase the overall height of the header while restricting separation of liquid-phase within the header under gravity.

[0080] While the subject disclosure has been described with reference to exemplary embodiments, various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the subject disclosure as defined by the appended claims.

Modifications may be made to adopt a particular situation or material to the teachings of the subject

disclosure without departing from the scope thereof. Therefore, it is intended that the subject disclosure not be limited to the particular embodiment disclosed, but that the subject disclosure includes all embodiments falling within the scope of the subject disclosure as defined by the appended claims.

[0081] In interpreting the specification, all terms should be interpreted in the broadest possible manner consistent with the context. In particular, the terms “comprises” and “comprising” should be interpreted as referring to elements, components, or steps in a non-exclusive manner, indicating that the referenced elements, components, or steps may be present, or utilized, or combined with other elements, components, or steps that are not expressly referenced. Where the specification claims refer to at least one of something selected from the group consisting of A, B, C . . . and N, the text should be interpreted as requiring only one element from the group, not A plus N, or B plus N, etc.

Claims

1. A laminated header for a microchannel heat exchanger, the laminated header comprising: a distribution plate comprising a first cut-out section of a first shape extending along a longitudinal axis of the distribution plate, wherein a width of the first cut-out section at least partially decreases along the longitudinal axis; a hole plate comprising a plurality of bore-holes formed therein and separated by a distance therebetween; a section plate comprising a plurality of second cut-out sections of a second shape different from the first shape; and wherein the hole plate is parallelly stacked between the distribution plate and the section plate.
2. The laminated header of claim 1, further comprising an inlet plate having an inlet bore-hole defined at one end of the inlet plate, and wherein the width of the first cut-out section decreases from a bottom end towards a top end of the distribution plate.
3. The laminated header of claim 1, further comprising an inlet plate having an inlet bore-hole defined at a middle portion of the inlet plate, and wherein the width of the first cut-out section decreases from the middle portion towards a bottom end and a top end of the distribution plate.
4. The laminated header of claim 1, further comprising an inlet plate having an inlet bore-hole defined at a top end of the inlet plate, and wherein the width of the first cut-out section decreases from a top end towards a bottom end of the distribution plate.
5. The laminated header of claim 1, further comprising a tube plate comprising a plurality of slots, wherein the laminated header is configured to receive and secure, within the plurality of slots, an end of a plurality of microchannel tubes of the microchannel heat exchanger.
6. The laminated header of claim 1, wherein the distribution plate, the hole plate, and the section plate are parallelly stacked between an inlet plate and a tube plate, and are brazed together to form the laminated header, and wherein an end of a plurality of microchannel tubes of the microchannel heat exchanger is inserted within a plurality of slots of the tube plate and brazed to create a leak-proof connection between the laminated header and the plurality of microchannel tubes.
7. The laminated header of claim 6, wherein the laminated header forms a fluidic passage extends from an inlet bore-hole towards the plurality of slots while extending in a first direction along the longitudinal axis, and further extending from the first cut-out section into each of the second cut-out sections and the plurality of slots, via the plurality of bore-holes, in a second direction perpendicular to the first direction and extending towards the plurality of slots or the microchannel tubes.
8. The laminated header of claim 1, wherein the first cut-out section has a trapezoidal shape with the width decreasing from a bottom end towards a top end of the distribution plate.
9. The laminated header of claim 1, wherein size or radii of the plurality of bore-holes of the hole plate increases in the direction away from an inlet bore-hole defined on an inlet plate, along the longitudinal axis.

- 10.** The laminated header of claim 1, wherein a constriction is defined on the first cut-out section of the distribution plate.
- 11.** The laminated header of claim 1, wherein the plurality of second cut-out sections associated with the section plate has a rectangular or square profile.
- 12.** The laminated header of claim 1, wherein size of the plurality of second cut-out sections increases in at least one direction along the longitudinal axis.
- 13.** The laminated header of claim 1, further comprising an inlet plate having an inlet bore-hole, and wherein the distribution plate has a thickness that is greater than a thickness of each of the inlet plate defining the inlet bore-hole and the hole plate.
- 14.** The laminated header of claim 1, wherein the laminated header is configured to allow a fluid from an inlet bore-hole defined on an inlet plate to flow within the first cut-out section of the distribution plate in along the longitudinal axis, between a top end and a bottom end, of the distribution plate.
- 15.** The laminated header of claim 14, wherein the laminated header is further configured to allow uniform flow of the fluid from the first cut-out section into each of the second cut-out sections of the section plate, via the plurality of bore-holes of the hole plate, in a second direction perpendicular to the first direction and extending towards the plurality of microchannel tubes, and wherein the laminated header further allows uniform flow of the fluid from the plurality of second cut-out sections into one or more ports associated with each of the microchannel tubes via a plurality of slots of a tube plate.
- 16.** The laminated header of claim 1, wherein adjacent second cut-out sections among the plurality of second cut-out sections of the section plate are separated by a baffle extending in a direction, orthogonal to the first direction and the second direction, along a width of the section plate.
- 17.** The laminated header of claim 1, wherein the distribution plate, the hole plate, and the section plate are parallelly stacked between an inlet plate and a tube plate, and wherein, an inlet bore-hole of the inlet plate is formed at a corresponding widthwise center of the inlet plate; the plurality of bore-holes are formed at a corresponding widthwise center of the hole plate; and a plurality of slots of the tube plate are formed at a corresponding width-wise center of a width of a tube plate.
- 18.** A heat exchanger, comprising: one or more laminated headers according to claim 1, wherein the one or more laminated headers are coaxially and sequentially stacked in a longitudinal direction to form a vertical header having one or more compartments, wherein the plurality of microchannel tubes associated with the microchannel heat exchanger is fluidically connected to the plurality of slots associated with the one or more laminated headers, and wherein the heat exchanger further comprises an external distributor comprising: an inlet configured to be fluidically connected to a supply tube; and one or more outlets, each configured to be fluidically connected to one of the laminated headers via feeder tubes or tube stubs.
- 19.** A laminated header for a microchannel heat exchanger, the laminated header comprising: a distribution plate comprising a first cut-out section of a first shape; a hole plate comprising a plurality of bore-holes formed therein and separated by a distance therebetween, a diameter of each of the plurality of bore-holes varies along a longitudinal axis of the hole plate; a section plate comprising a plurality of second cut-out sections of a second shape different from the first shape; and wherein the hole plate is parallelly stacked between the distribution plate and the section plate.
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